

Foundations of Artificial Intelligence

M. Helmert
C. Büchner

University of Basel
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Exercise Sheet 8

Due: April 26, 2026

Please note the submission rules at the end of this exercise sheet. Non-adherence to these rules may lead to your submission not being marked.

Exercise 8.1 (1+2 marks)

Consider the propositional formula $\varphi = ((A \vee \neg C) \rightarrow (\neg C \wedge (\neg A \vee B)))$.

- (a) Is φ satisfiable, unsatisfiable, falsifiable, valid? Briefly justify your answers.
- (b) Transform φ into CNF using the equivalences from Chapter E2. Provide all intermediate formulas that result from applying an equivalence transformation.

Exercise 8.2 (0.5+1+2.5 marks)

- (a) Let φ and ψ be propositional logic formulas. If φ has 4 models and ψ has 3 models, what is the smallest and what is the largest number of models that $(\varphi \vee \psi)$ can have?
- (b) State a propositional logic formula over $\Sigma = \{A, B, C\}$ that expresses that exactly one variable in Σ is true.
- (c) Show with resolution that the CNF formula

$$(A \vee \neg B) \wedge (A \vee B \vee \neg C) \wedge (\neg A \vee \neg B \vee E) \wedge (B \vee C) \wedge (B \vee \neg C) \wedge (\neg B \vee D) \wedge \neg E$$

is unsatisfiable.

Note: This exercise was part of an exam question in 2024.

Exercise 8.3 (2+1 marks)

- (a) Perform DPLL on

$$\{\{\neg A, C\}, \{A, B, E\}, \{\neg B, \neg D\}, \{\neg B, \neg E\}, \{\neg C, D, E\}, \{B, \neg C, E\}, \{B, \neg E\}\}$$

choosing variables alphabetically (also for unit propagation if there are several choices) and always assigning **T** first for splitting. Provide the application of all rules and the resulting tree as done in the lecture (chapter E4). What is the result of the algorithm?

- (b) Would always assigning **F** first result in a smaller tree? Justify your answer.

Submission rules:

- Submit one solution in groups of up to two students, stating the names of all group members in the submission. All group members are equally responsible for all submitted exercises.
- Create a single A4 PDF file for all non-programming exercises. Solutions in L^AT_EX are preferred, but clearly legible scans in A4 with reasonable contrast and file size are accepted as well.

- In case there are programming exercises, upload a ZIP file (ending .zip, .tar.gz or .tgz; *not* .rar or anything else) including your PDF and those code textfiles required by the exercise. Put your names in a comment on top of each code file. Make sure your code compiles and test it. Code that does not compile or which we cannot successfully execute will not be graded.
- Include a statement on scientific integrity at the end of your submission (see below).

Statement on scientific integrity:

- All submitted solutions must be the product of your own work.
- If your written solutions were produced using any help (e.g., from peers or a large language model), list in what way these people or tools contributed to the solutions.
- We do not mark submissions that lack a statement on scientific integrity.
- Examples:
 - “We solved all exercises on our own.”
 - “We used ChatGPT to explain exercise 7 which we did not understand.”
 - “We had no time to solve this exercise sheet and copied all solutions from our friends.”
(If this is the case, thanks for your honesty but this is plagiarism, results in a warning and no marks. It is preferable to submit no solution at all.)